

Ankit Sarkar

+91-9614161186 | sarkarankit123456@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

Indian Institute of Information Technology Guwahati (IIITG)
B.Tech in Computer Science and Engineering

Guwahati, India
Aug 2023 – Jun 2027

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, Tailwind CSS, Radix UI (Shadcn), Framer Motion, HTML5, CSS3

Backend & Database: Node.js, Express.js, REST APIs, PostgreSQL, Supabase (Auth, Storage, Row-Level Security), Zod

Tools & Platforms: Git, GitHub, VS Code, Figma, Docker, Vercel

Practices: Agile/Scrum, Clean Code, Web Accessibility, SEO Optimization, Software Testing

Projects

Arthix – Personal Finance & Group Expense Manager

[Live Demo](#) | [GitHub](#)

React 19, Next.js 16, TypeScript, Supabase, PostgreSQL, Tailwind CSS

- Architected a multi-tenant finance platform with granular React Context state caching, reducing redundant network requests and unnecessary UI re-renders.
- Enforced strict data isolation with PostgreSQL Row-Level Security policies and type-safe Server Actions validated end-to-end with Zod.
- Built a client-side CSV/XLSX bank-statement parsing pipeline that auto-detects major Indian banks (PNB, HDFC, SBI, ICICI), cleans transaction data, and auto-categorizes spending.
- Designed a collaborative expense-splitting system with 5 split modes and a debt-simplification algorithm that minimizes settlement transactions.

UniAdmin – University Administration & Business Process Automation Platform

[GitHub](#)

React 19, TypeScript, Node.js, Express, Supabase, PostgreSQL

- Built a multi-role SaaS platform (Student / Faculty / Admin) automating enrollment, grading, attendance, leave approvals, and exam re-evaluation workflows.
- Designed a layered, modular-monolith architecture separating presentation, business logic, and a Supabase-backed data access layer.
- Built a secure Node.js/Express API gateway using the Supabase Admin SDK for safe user onboarding without exposing credentials client-side.
- Implemented a 20+ table PostgreSQL schema with Row-Level Security policies and a secure OTP-based password recovery flow; authored white-box and black-box test plans.

MLP Optimization & Regularization Study (CS300 Academic Project)

[Report](#)

Python

- Analyzed and benchmarked optimization and regularization techniques for Multi-Layer Perceptrons on the Wine Quality and California Housing datasets.
- Evaluated multiple optimizers and regularization techniques, documenting experimental findings and performance comparisons in a technical report.

Personal Portfolio Website

[Live Demo](#) | [GitHub](#)

Next.js, React, Tailwind CSS

- Designed and built a personal portfolio site showcasing skills, projects, and experience with a modern, responsive UI.

MoodMap – Cross-Platform Mental Wellness App (In Progress)

[GitHub](#)

React Native (Expo), TypeScript, SQLite, Supabase, Spotify API

- Building a full-stack mobile wellness app with mood tracking, journaling, and a 15+ visualization analytics dashboard on an offline-first SQLite + Drizzle ORM architecture.
- Engineering a 3-layer music recommendation engine combining rule-based mapping, local listening history, and Spotify Web API search for context-aware playback.
- Integrating Spotify OAuth 2.0 PKCE authentication with race-condition-safe token refresh, plus biometric app lock via expo-local-authentication.

Leadership & Achievements

Coordinator, Robotics Club (Technocrats)

Guwahati, India

Technical Board, IIIT Guwahati

- Led and coordinated robotics events including the Space Day Rover Competition and Entropy FastTrack Robo Car Racing Competition, managing volunteers, logistics, and participant support.
- Gained hands-on experience with ESP32, sensor integration, wireless communication, and circuit design while building and troubleshooting RC robots.